Md Nahidul Islam

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Research website | Google Scholar | ORCID | GitHub | LinkedIn

Research Interests

Autonomous driving and intelligent transportation; safety-aware motion planning and model compression; cybersecurity and anomaly detection in connected cyber-physical systems; robust LiDAR perception; trajectory forecasting and interaction modeling; trustworthy evaluation of autonomous decisions; traffic-aware vehicle–drone logistics.

Education

Ph.D. in Computer Science *Jan. 2024 – Present*
UNIVERSITY OF MEMPHIS Memphis, TN, USA

Advisor: Prof. Myounggyu Won.

M.S. in Computer Science *May 2026*
UNIVERSITY OF MEMPHIS Memphis, TN, USA

GPA: 3.95/4.00 | Thesis: *BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning*.

B.Sc. in Computer Science and Engineering *2012 – 2017*
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY (BUET) Dhaka, Bangladesh

University Research Appointments

Graduate Research Assistant — Transportation and Route Modeling *Summer 2026*
UNIVERSITY OF MEMPHIS; SUPERVISOR: DR. MYOUNGGYU WON Memphis, TN

- Developed methods for transferable route search and travel-time estimation across unfamiliar and evolving road networks.
- Investigated route-search efficiency, cross-city generalization, and historically accurate road-network evaluation, contributing to the TRaCE and RIVET research manuscripts.

Graduate Research Assistant — ChemXtract (NIH-Funded Interdisciplinary Research) *May – Aug. 2025*
UNIVERSITY OF MEMPHIS; SUPERVISOR: DR. QIANYI CHENG Memphis, TN

- Developed an automated image-analysis framework for extracting and interpreting chemical structure information from scientific images.
- Designed and evaluated machine-learning-based classification and image-processing components to support chemical research workflows and reduce manual analysis.

Student Research Assistant — DHS/FEMA-CTG 23 Cybersecurity Project *Jul. – Aug. 2024*
UNIVERSITY OF MEMPHIS; SUPERVISOR: DR. MOHD HASAN ALI Memphis, TN

- Contributed to the DHS/FEMA-supported project "Cybersecurity Issues with Operational Technology and Distributed Energy Resources."
- Developed cybersecurity training materials covering threats to operational technology and distributed energy resources, including attack scenarios, anomaly-detection concepts, and mitigation techniques for critical infrastructure.

Graduate Research Assistant — NSA-Funded Project; Vehicle–Drone Transportation *Jun. 2024*
UNIVERSITY OF MEMPHIS; SUPERVISOR: DR. MYOUNGGYU WON Memphis, TN

- Developed cybersecurity course materials for high-school students using autonomous radio-controlled (R/C) cars as a hands-on learning platform, under an NSA-funded subaward (H98230-21-1-0320).
- Researched traffic-aware vehicle–drone coordination for last-mile delivery, integrating machine-learning-based travel-time prediction with vehicle–drone route optimization.
- Contributed to a simulation framework incorporating dynamic traffic conditions, delivery constraints, and operating costs to evaluate reliable and efficient transportation strategies.

Professional Experience

Graduate Assistant — Teaching Support *Jan. 2024 – Present*
UNIVERSITY OF MEMPHIS Memphis, TN

- Supported *Operating Systems* and *Capstone Software Project* teaching activities, including grading, project assessment, and student mentoring; doctoral research is conducted separately through credit-bearing study.

Technical Mentor and Volunteer *2024 – Present*

CYBER AMBASSADORS TECH CAMP (AUTONOMOUS R/C CAR CYBERSECURITY) Memphis, TN

- Mentored high-school students in introductory programming, cybersecurity, and autonomous R/C vehicle demonstrations, with continued participation in subsequent camps.

Assistant Director (ICT) *Nov. 2020 – Jan. 2024*
BANGLADESH BANK (CENTRAL BANK OF BANGLADESH) Dhaka, Bangladesh

- *Enterprise Data Warehouse Wing:* Developed and maintained ETL workflows and web applications/REST APIs for regulatory financial data collection, validation, and reporting.
- *Network Security Unit:* Monitored firewall and IDS/IPS events, evaluated vulnerabilities, assisted with security-incident investigation, and reviewed network access controls.
- *Certifying Authority Unit:* Supported PKI operations, including digital certificate issuance, renewal, revocation, and cryptographic key-security procedures.

Software Engineer *Feb. 2020 – Nov. 2020*
BINATE SOLUTIONS LTD. Dhaka, Bangladesh

- Developed REST APIs, machine-learning services, and cloud-enabled applications using Django, TensorFlow, and database technologies.

Research Assistant *Aug. 2018 – Jan. 2020*
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY (BUET) Dhaka, Bangladesh

- Conducted research in low-resource speech recognition and language processing; contributed to peer-reviewed papers at ICASSP, LREC, and NAACL.

Chief Technology Officer *Apr. 2018 – Jul. 2018*
CHIRON SOLUTIONS LTD. Dhaka, Bangladesh

- Oversaw the design, development, and deployment of technology solutions to improve business processes and efficiency, and led the technology team in planning and delivering projects.

Information Analyst, MIS *Feb. 2017 – Mar. 2018*
ACI LTD. Dhaka, Bangladesh

- Built reports and analytics tools supporting business operations and management information systems.

Publications, Thesis, and Research Manuscripts

*CV author indicated in **bold**; published, accepted, submitted, and planned work are distinguished below.*

Published and accepted peer-reviewed papers

[1] L. Won, N. H. Nguyen, **M. N. Islam**, L. Das, and M. Won, "LiDARBridge: Range-Dependent LiDAR Noise Modeling and Prototype Alignment for Simulation-to-Real 3D Object Detection," *accepted at IEEE Consumer Communications & Networking Conference (CCNC)*, 2027.

[2] L. Hwang, **M. N. Islam**, M. H. Ali, D. Dasgupta, and M. Won, "Detecting Stealthy Anomalies with Autoencoders Using Windowed Variance-Aware Loss Functions," *Proc. IEEE Military Communications Conference (MILCOM)*, 2025. [DOI]

[3] N. Sadeq, S. Ahmed, S. S. Shubha, **M. N. Islam**, and M. A. Adnan, "Bangla Voice Command Recognition in End-to-End System Using Topic Modeling Based Contextual Rescoring," *Proc. IEEE ICASSP*, 2020.

[4] S. Ahmed, N. Sadeq, S. S. Shubha, **M. N. Islam**, and M. A. Adnan, "Preparation of Bangla Speech Corpus from Publicly Available Audio & Text," *Proc. LREC*, 2020, pp. 6586–6592.

[5] S. S. Shubha, N. Sadeq, S. Ahmed, **M. N. Islam**, M. A. Adnan, M. Y. A. Khan, and M. Z. Islam, "Customizing Grapheme-to-Phoneme System for Non-Trivial Transcription Problems in Bangla Language," *Proc. NAACL-HLT*, 2019, pp. 3192–3201.

Completed master's thesis

[6] **M. N. Islam**, "BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning," M.S. thesis, University of Memphis, 2026. [thesis]

Manuscripts submitted for peer review (decisions pending)

[7] **M. N. Islam** and M. Won, "TRaCE: Transferable Route Search and Travel-Time Estimation on Unseen Road Networks," submitted to *The ACM Web Conference (WWW)*, 2027.

[8] **M. N. Islam** and M. Won, "RIVET: Revisiting Cross-City Travel-Time Estimation as Road Networks Evolve," submitted to *The ACM Web Conference (WWW)*, 2027.

[9] **M. N. Islam** and M. Won, "What Aggregate Metrics Miss: A Residual-Signal Protocol for Interaction Representations in Trajectory Forecasting," submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.

[10] L. Won, **M. N. Islam**, and M. Won, "ATLAS: Adaptive Temporal Weighting and Location Optimization for Multi-Frame Attacks

in Autonomous Driving," submitted to *IEEE International Conference on Communications (ICC), CISS Symposium*, 2027.

Manuscripts in preparation / planned (not submitted)

[11] **M. N. Islam** and M. Won, "CRiC: Protocol-Scoped Counterfactual Risk Certification for Interactive Driving," in preparation for *CVPR*, 2027.

[12] **M. N. Islam**, M. H. Ali, D. Dasgupta, and M. Won, "BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning," journal manuscript planned for *IEEE Transactions on Intelligent Vehicles*.

Research Projects

DyadCode: Path-Aware Interaction Representation Learning *2026–Present*
ONGOING RESEARCH

- Developing selectively gated, path-aware representations of interactions between road users, with model-level tests and evaluation of downstream trajectory prediction on AV2 and WOMD.

CRiC: Counterfactual Risk Certification for Interactive Driving *2026–Present*
IN PREPARATION FOR CVPR 2027

- Investigating risk estimates for alternative driving actions, the limits of confidence in those estimates, and decision-abstention mechanisms for unsupported safety-critical choices.

RSP: Residual-Signal Protocol for Trajectory Forecasting *2026–Present*
SUBMITTED TO IEEE ICRA 2027

- Developed evaluation tests for whether learned representations encode meaningful interactions between road users beyond information available from basic scene and motion factors; studied across forecasting models and datasets.

TRaCE: Transferable Route Search on Unseen Road Networks *2026–Present*
SUBMITTED TO THE ACM WEB CONFERENCE (WWW) 2027

- Designed methods for efficient alternative-route search in unfamiliar cities; reported 64.8–73.8% fewer search expansions and 2.03–2.52× faster route generation in specified held-out-city evaluations.

RIVET: Travel-Time Evaluation as Road Networks Evolve *2026–Present*
SUBMITTED TO THE ACM WEB CONFERENCE (WWW) 2027

- Developed historically accurate road-network evaluation to prevent leakage from modern maps; found 3.7–5.1% of present-day-routable queries were not routable on corresponding historical maps in the evaluated settings.

ATLAS: Multi-Frame Attacks in Autonomous Driving *2026*
SUBMITTED TO IEEE ICC 2027 (CISS SYMPOSIUM)

- Studied temporally coordinated adversarial attacks against autonomous-driving systems; collaborative manuscript undergoing peer review.

LiDARBridge: Simulation-to-Real 3D Object Detection *2025–2026*
IEEE CCNC 2027 — ACCEPTED

- Contributed to range-dependent LiDAR noise modeling and prototype alignment to reduce discrepancies between simulated and real-world point clouds for autonomous-vehicle object detection.

Stealthy Cyber-Anomaly Detection with Windowed Autoencoders *2024–2025*
IEEE MILCOM 2025 — PUBLISHED

- Co-designed and experimentally evaluated windowed standard-deviation-aware reconstruction losses (WSR/WISR) to detect low-visibility anomalous network activity. Evaluated attack-class detection against standard loss-function baselines on CIC-IDS2017.

ChemXtract: Scientific Image-Analysis Research Tool *Summer 2025*
INTERDISCIPLINARY RESEARCH UNDER PROF. QIANYI CHENG

- Developed and evaluated an image-analysis classifier as part of a University of Memphis NIH-supported research program; provided software and documentation for subsequent laboratory use.

BucketKD: Safety-Aware Motion-Planner Compression *Fall 2024–Present*
M.S. THESIS; JOURNAL MANUSCRIPT PLANNED FOR IEEE TRANSACTIONS ON INTELLIGENT VEHICLES

- Designed planning-semantic bucket-based feature distillation and relative-motion-aware waypoint distillation to retain planning-critical distinctions under model compression.
- In a same-capacity 16.6M-parameter CARLA/Bench2Drive student comparison, achieved Driving Score 37.26 versus 30.57 for PlanKD, and 0.678 versus 0.748 collisions/km.

Professional Projects

Portal for Data Collection *2021–2023*
Web application — financial data ingestion and reporting

- Developed a portal through which users uploaded CSV files within scheduled reporting windows; implemented rule-based checks to validate incoming data before storage.
- Built reporting features, interactive dashboards, and administrative user-management functionality; integrated the frontend with Oracle-backed data services and containerized the application for deployment.
- **Languages:** Python, JavaScript; **Frameworks:** Django, React, Redux; **Database/Tools:** Oracle, Docker, PyCharm.

Automated Chatbot Using Rasa *2020*
Conversational customer-service application

- Developed a chatbot to answer frequently asked questions, deliver reference information and questionnaires, and provide project-booking assistance.
- Implemented conversational flows for appointment scheduling with sales representatives, combining dialogue handling with backend application services.
- **Language:** Python; **Frameworks:** Rasa, TensorFlow, Django; **Database/Tools:** PostgreSQL, PyCharm.

Digital Building Safety *2020*
Building-document and compliance management

- Developed a system for organizing and retrieving building-related documents and tracking their expiration dates; automated email reminders for upcoming renewals.
- Implemented document-type management, a calendar for deadline tracking, and dashboards with charts to support compliance monitoring and administrative decisions.
- **Languages:** Python, JavaScript; **Frameworks:** Django, AngularJS, Bootstrap; **Database/Tools:** PostgreSQL, PyCharm, VS Code.

Principal Component Analysis of Text Status and Stock Market *2019*
Text analytics and financial data exploration

- Built an application to examine Twitter text using principal component analysis (PCA), identifying prominent patterns and terms in social-media content.
- Applied dimensionality-reduction techniques to stock-market information to support pattern exploration and data-driven comparison of financial indicators.
- **Languages:** Python, JavaScript; **Frameworks/Libraries:** Django, jQuery, Bootstrap, PCA; **Database/Tools:** PostgreSQL, PyCharm.

Digital Paper Commercialization *2018–2019*
Permissioned blockchain application

- Contributed to a permissioned platform for authorized organizations to issue, transfer, and redeem digital commercial paper.
- Supported transaction workflows for buyers and issuers, including recording ownership transfers and redemption at maturity.
- **Language:** JavaScript; **Frameworks/Tools:** Node.js, Hyperledger Fabric, blockchain, Visual Studio Code.

Financial Data Portal *2018*
Financial-information and visualization platform

- Developed a portal aggregating company financial statements, valuation data, financial ratios, and comparative company information.
- Integrated market-news views and interactive TradingView charts to support technical analysis, competitor comparison, and financial-data exploration.
- **Languages:** Python, JavaScript; **Frameworks/Libraries:** Django, jQuery, Bootstrap; **Database/Tools:** PostgreSQL, TradingView, PyCharm.

Customer Fraud Detection Using Neural Networks *2017–2018*
Machine-learning application

- Built and trained an LSTM sequence-to-sequence neural-network model for identifying potentially fraudulent customer activity using an internal organizational dataset.
- Evaluated the model on the organization's data; the original project record reports **92.87% classification accuracy**.
- **Languages:** Python, JavaScript; **Frameworks/Tools:** Django, Scrapy, jQuery, Bootstrap, PostgreSQL, PyCharm; **Model:** LSTM sequence-to-sequence.

Churn Analysis and Prediction Using Neural Networks *2017–2018*
Customer analytics and prediction

- Developed an LSTM sequence-to-sequence model to analyze patterns in customer activity and predict churn using an internal organizational dataset.
- Trained and evaluated the prediction system on company data; the original project record reports **89.63% accuracy**.

- **Languages:** Python, JavaScript; **Frameworks/Tools:** Django, jQuery, Bootstrap, PostgreSQL, PyCharm; **Model:** LSTM sequence-to-sequence.

Automated Chatbot with Speech *2017*
Bilingual conversational application

- Developed a sequence-to-sequence chatbot using an internal dataset, with Bangla/English language recognition and language-appropriate responses.
- Integrated weather-service APIs for location-specific updates; the original project record reports **73.59% accuracy** for the model.
- **Languages:** Python, JavaScript; **Frameworks/Tools:** Django, jQuery, Bootstrap, PostgreSQL, PyCharm, weather APIs; **Model:** LSTM sequence-to-sequence.

Sentiment Analysis *2017*
Social-media and news text classification

- Collected text from Facebook posts, Twitter posts, and news articles through web-crawling and data-ingestion workflows.
- Classified content as positive, negative, or neutral using natural-language-processing libraries and services; compared analyses across multiple available tools.
- **Languages:** Python, JavaScript; **Frameworks/Tools:** Django, Scrapy, jQuery, Bootstrap, PostgreSQL, Google Natural Language API, IBM Watson NLU, Azure NLP, NLTK, TextBlob, VADER.

E-Commerce Website for Crop Care and Public Health (ACI) *2017*
Commerce platform with government integration

- Developed purchase workflows for both general customers and government customers in a crop-care and public-health e-commerce platform.
- Implemented API-based exchange of government-customer information and purchase records to support reporting and regulatory data flows.
- **Languages:** Python, JavaScript; **Frameworks/Libraries:** Django, jQuery, Bootstrap; **Database/Tools:** PostgreSQL, PyCharm.

Motors Insight *2017*
Branch sales and service monitoring

- Built an internal application for staff across multiple branches to monitor daily sales activities and customer interactions.
- Organized operational information to support branch-level reporting and timely customer-service follow-up.
- **Languages:** PHP, JavaScript; **Frameworks/Libraries:** CodeIgniter, jQuery, Bootstrap; **Database/Tools:** Microsoft SQL Server, NetBeans.

Business Summary *2017*
Mobile business-intelligence dashboard

- Developed a mobile dashboard displaying sales, revenue growth, and gross-profit metrics for business-performance monitoring.
- Added historical-period comparisons and interactive charts to help users interpret trends and changes in key indicators.
- **Language:** C#; **Frameworks/Tools:** Xamarin, ASP.NET, Visual Studio.

Fault Tracking System *2016–2017*
Incident tracking and reporting

- Developed a role-based system for authorized officers to document rule violations and take actions consistent with organizational procedures.
- Created incident-record workflows and reports for monitoring and follow-up in restricted areas.
- **Languages:** C#, JavaScript; **Frameworks/Libraries:** ASP.NET, jQuery; **Architecture/Tools:** Three-tier architecture, Crystal Reports, Microsoft SQL Server, Visual Studio.

Inventory Management System *2016–2017*
Desktop inventory and pricing application

- Developed a desktop application to manage inventory items and their components, configurable attributes, and branch-specific pricing.
- Implemented administrative editing of items and rates, together with detailed inventory reports and year-over-year comparisons.
- **Language:** C#; **Frameworks/Patterns:** WPF, .NET Framework, MVVM, dependency injection, Entity Framework (Code First); **Tools:** Messenger service, Crystal Reports, Microsoft SQL Server, Visual Studio.

Awards and Honors

- **College of Arts and Sciences Travel Enrichment Award**, University of Memphis, Spring 2027 ($500 research-travel support for IEEE CCNC participation).
- **1st Place, Individual Category**, Tech901 CTF, 2025.
- **1st Position, Team Category**, Financial Institutions Cyber Drill, organized by BGD e-GOV CIRT, ICT Division, Bangladesh, 2021.
- **Letter of Appreciation**, awarded by the Governor of Bangladesh Bank, 2021.

- **Technical Scholarship**, Bangladesh University of Engineering and Technology, 2012.
- **General Scholarship**, Secondary School Certificate, 2009.
- **Talent Pool Scholarship**, High School, 2007.

Research Presentations and Outreach

- **FLEET Industry Technology Forum, Graduate Research Showcase (May 7, 2026):** Presented BucketKD at Dallara U.S. Headquarters in Speedway, Indiana, to participants of an NSF-supported industry–university research center. **(NSF travel support)**

Technical Skills

Programming / Web Python, C++, C, C#, Java, JavaScript, SQL/PL-SQL, R; Django, FastAPI, React, Node.js, ASP.NET

AI / Robotics PyTorch, TensorFlow, Hugging Face, OpenCV; CARLA, Donkey Car, motion planning, trajectory forecasting, AV2, WOMD

Systems / Data Linux, Git, Docker, HPC/GPU clusters, AWS, CI/CD, Oracle, PostgreSQL, MySQL, MS SQL Server